Numerical Formulation & Setup

Solver

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

$$\nabla^2 p = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*$$

- Basilisk centered: CUDA, fp32, fixed grid
- incflo/AMReX: CUDA, AMR

Simulation Parameters

- Core radius: $\sigma = 0.09$
- Separation: $2d = 0.28$ $(D = 0.14)$
- Viscosity: $\nu = 10^{-5}$ $(Re_\Gamma \approx 2.5 \times 10^3)$
- Grid levels: $L = 4 \ldots 8$ $(64^3 \rightarrow 256 \times 256 \times 1024)$
- CFL: 0.3–0.5
- AMR: ratio 4, VORTERR $0.3 \ldots 10^{-2}$

Fixed-Grid GPU: Cost Model

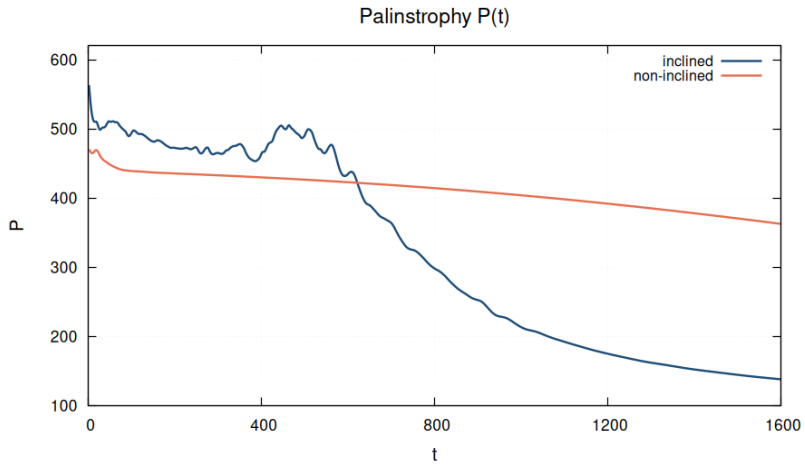
Level	Grid	Cells	VRAM	s/step
L6	$64 \times 64 \times 256$	1.05M	0.15 GB	0.2
L7	$128 \times 128 \times 512$	8.4M	1.1 GB	1.6
L8	$256 \times 256 \times 1024$	67.1M	7.5 GB	2.8

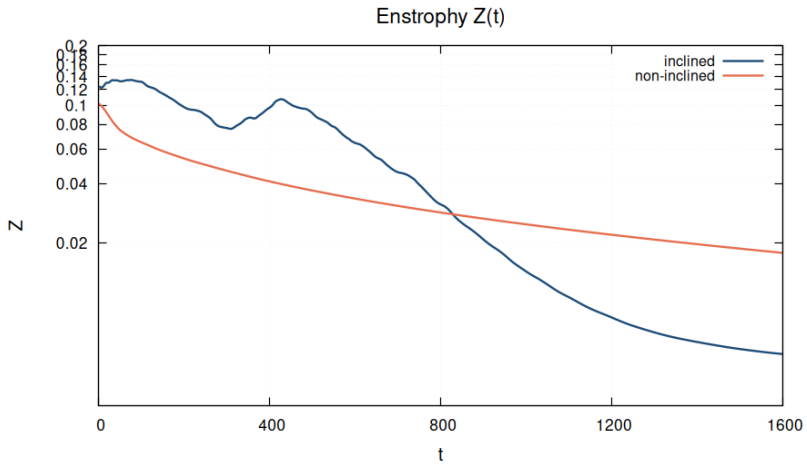
- fp32: 4 bytes per value
- $\approx$ 120 bytes/cell (fields, ghosts, multigrid, CUDA context)
- 8 GB $\rightarrow$ 67M cells; 16 GB $\rightarrow \approx$140M
- L9 (537M cells) would need $\approx$58 GB

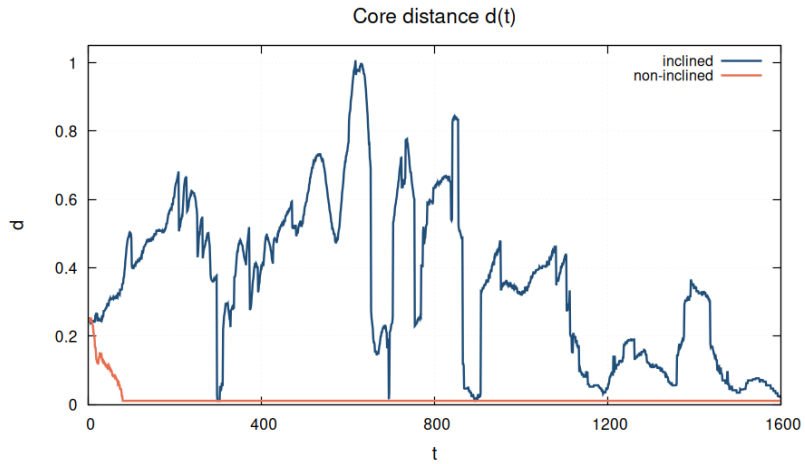
Conservation Laws & Dynamics Diagnostics

Conservation diagnostics at $T = 1600$ (L7)

	TILT=2	TILT=0
$\Delta\Gamma/\Gamma_0$	-42.9%	-7.8%
$I(T)/I_0 - 1$	$+0.038$	$+1.34$

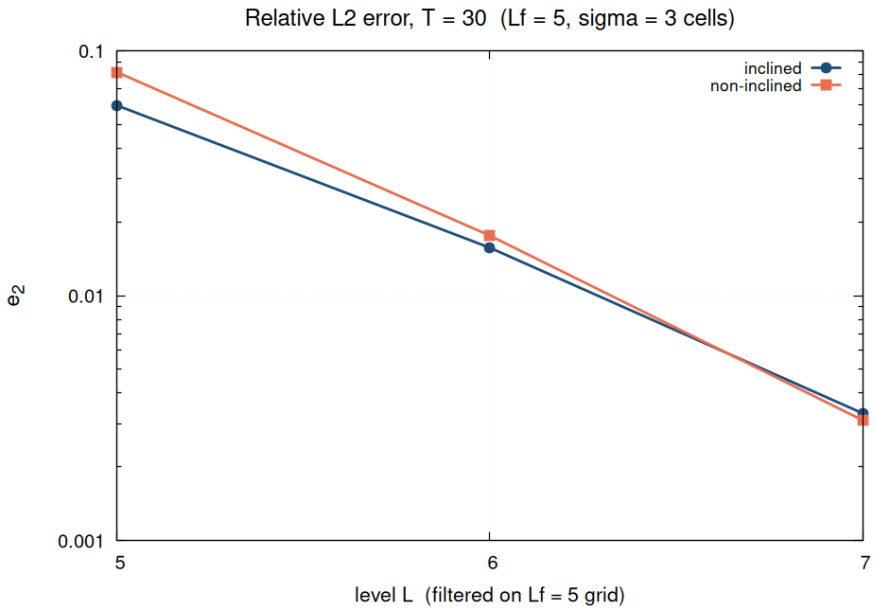


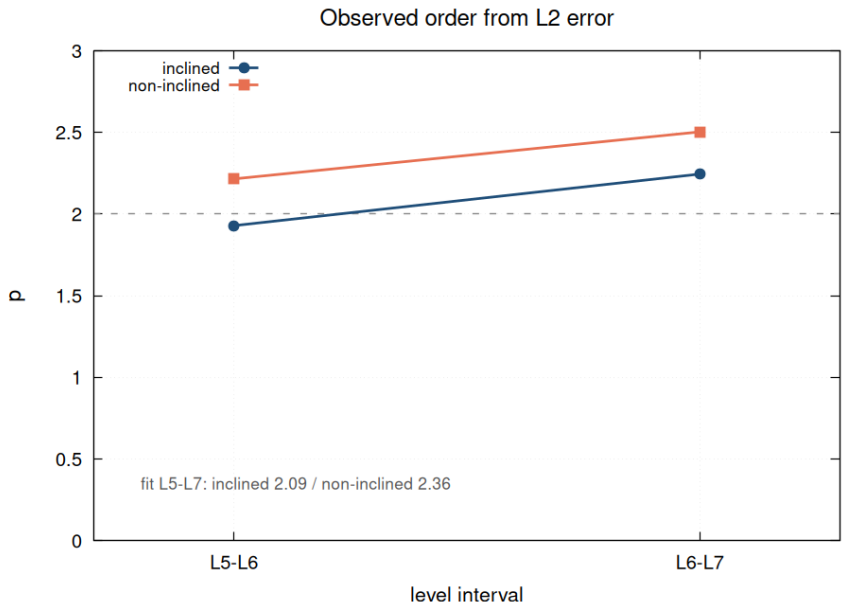




Γ: volume average; I: angular impulse; Z: enstrophy; P: palinstrophy; d: core distance.

Grid Convergence & Parameter Sensitivity





Poisson Solver: Elliptic Residuals

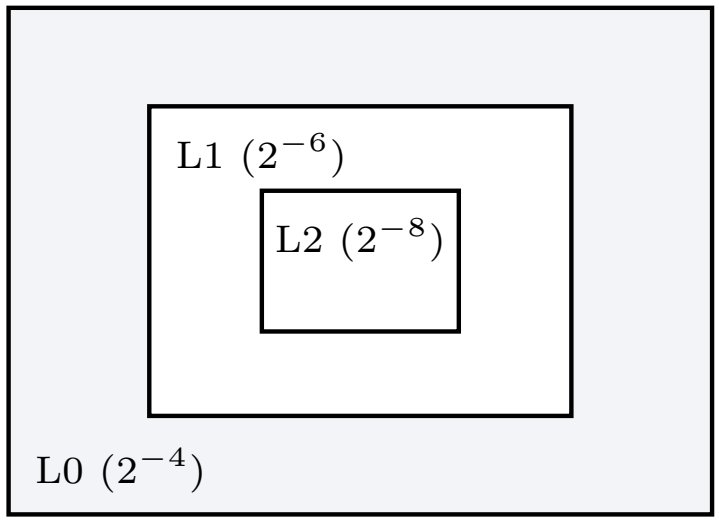
Max projection residual, $T = 30$

MAXLEVEL	inclined	non-incl.
4	7.6e-4	6.7e-4
5	3.2e-3	9.8e-4
6	8.0e-3	1.9e-3
7	1.1e-2	2.3e-3
8	1.3e-2	1.5e-3

- Residual grows with the level (more cells, same tolerance) but stays below 1.3×10^{-2} at L8
- Non-inclined case: below 2.3×10^{-3} at all levels
- Fixed-grid solver uses 2 V-cycles per step
- Residual is relative to $\max |\nabla \cdot \mathbf{u}^* / \Delta t|$

Adaptive Grid: How It Is Built

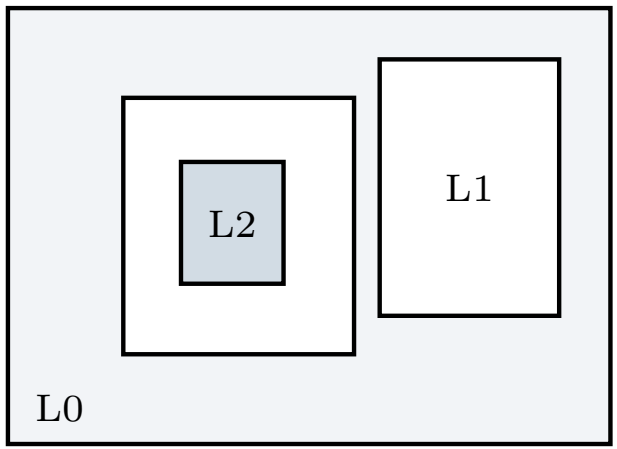
- Uniform base 2^{MINLEVEL} (e.g. $16 \times 16 \times 64$)
- Refinement ratio 4 per level; levels $= \lceil (\text{MAXLEVEL} - \text{MINLEVEL})/2 \rceil$
- Tagging: $|\boldsymbol{\omega}| >$ VORTERR (one threshold per level)
- Periodic regrid; average down/up between levels
- Cell budget (max_cells): if the hierarchy explodes, thresholds are raised



Nested refinement: detail only where $|\boldsymbol{\omega}|$ is large.

AMR on GPU: Data Structures & Regrid Cycle

- Each level is a *BoxArray*: a list of non-overlapping boxes covering only the regions that need refinement
- Data: one MultiFab per field ($\boldsymbol{\omega}$, $\mathbf{u}_f$, p, …) = BoxArray + distribution mapping + device arrays (fp32)
- Tagging: a GPU kernel computes $|\boldsymbol{\omega}|$ cell-by-cell and flags cells above VORTERR
- Clustering (Berger–Oliger): flagged cells are grouped into a few large boxes, to limit kernel launches
- Regrid: build the new fine BoxArray, average down the fine solution, fill coarse/fine ghosts



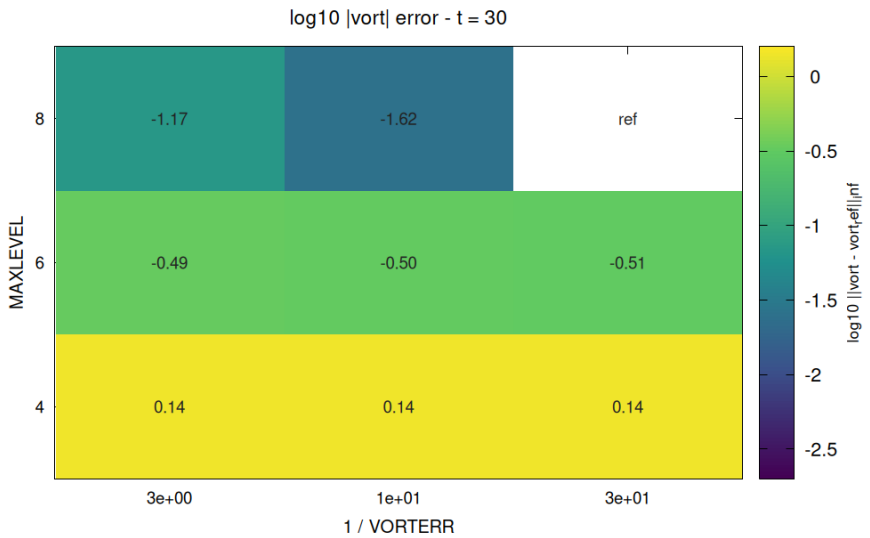
every level is a set of boxes; the fine data live only inside them

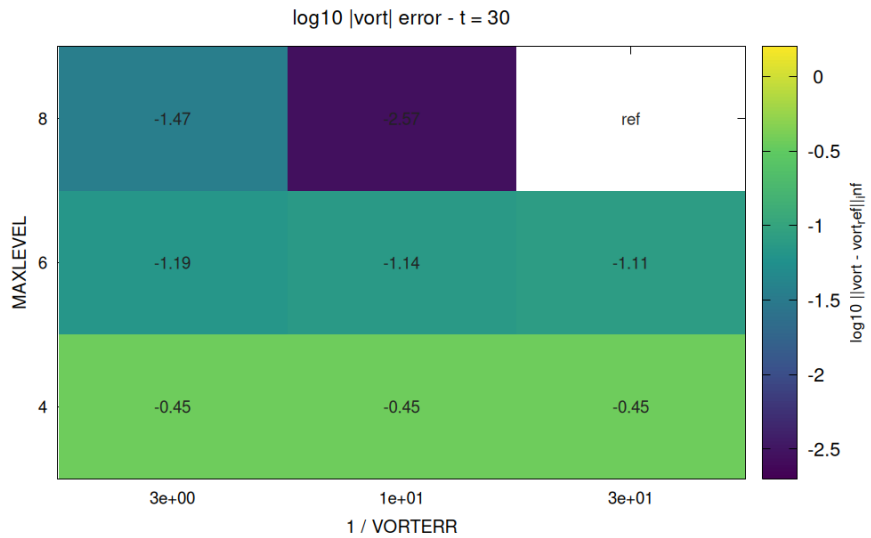
AMR on GPU: Execution Model & Costs

- Level update = sequence of GPU kernels: advection/diffusion are one kernel per box, while the Poisson solve uses a multigrid across levels
- Ghost cells: copy from same-level neighbours, interpolate from the coarser level; after each regrid, average down and fill patches
- Global reductions (norms, number of tagged cells) must complete before a regrid; kernels on a level run in order on the default stream
- Memory: ≈ 357 MB per million cells, about $3\times$ the fixed grid; the arena reuses buffers
- Measured: 11.7M cells $\rightarrow$ 6.4 GB; $T = 30$ cap: 15M cells (8 GB GPU)

	fixed	AMR
bytes/cell	$\sim$120 B	$\sim$357 MB/M
cells at 8 GB	67M	11–15M
strength	uniform	local

Adaptive Grid: Error Map at $T = 30$





Methods & Software

- Solvers: Basilisk (Popinet, 2015), centered scheme with CUDA backend; incflo / AMReX (Zhang et al., 2019) for block-structured AMR.
- Implementation: second-order Crank–Nicolson viscous term (`centered_cn.h`); on-the-fly palinstrophy P and angular impulse I; AMR error-map and GPU cost pipelines.
- Hardware: single NVIDIA RTX 3070 (8 GB), single precision; fixed-grid runs L4–L8 up to $T = 1600$.